DEPARTMENT OF CSE, University College of Engineering, O.U.
Software Engineering

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Class Test 2 — BE (AML) IV semester
Date: 27/06/2025; Test duration 1 ½ hours — Max marks: 30

Instructions: Set a margin of at least *2 in* on left side for each of the pages.

Part A (10 Marks) Each Question weighs 2 marks

1 a). Explain what is the main objective of component-based architecture?
1 b). What is the advantage of Object Constraint Language (OCL) over Unified Modelling Language (UML)?
1 c). What is the main objective of building the **Analysis Model** of a software product.
1 d). What are the typical solutions that an **Analyses Pattern** provides.
1 e). Illustrate the difference between Software Design Vs Software Architecture.

Part B (2 x 10 Marks)
Answer any 2 Questions.

2 a). How do you conduct Component – Level Design?
b). Explain what is a S/W Component? Describe the general Characteristics of Components.

3 a). What is User Interface? Describe all of the types of User Interface types.
b). With the help of the UML diagram, write the OCL construct for the following constraint. "The person who is younger than 18 years cannot have income of more than 5000". *[Assume the suitable properties and behavior of Person object.]*

4 a). Explain the purpose of first 3 levels of a given DFD diagram.
b). What do you understand by Flow oriented modeling.

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